Review of Last Lecture

Laplace Transform (LT)

For a signal $f(t)$, after multiplying by an attenuation factor $e^{-\sigma t}$(where σ is any real number), it easily satisfies the absolute integrability condition. According to the definition of Fourier transform:

$$F_1(\omega) = F\left[f(t)\cdot e^{-\sigma t}\right] = \int_{-\infty}^{+\infty}\left[f(t)e^{-\sigma t}\right]\cdot e^{-j\omega t}\,dt$$

$$= \int_{-\infty}^{+\infty} f(t)\cdot e^{-(\sigma + j\omega)t}\,dt = F(\sigma + j\omega)$$

Let $\sigma + j\omega = s$, which has the dimension of frequency and is called complex frequency.

ω is the oscillation frequency, σ controls the rate of attenuation

then $F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt$

$$f(t) \xleftrightarrow{LT} F(s)$$

4.2 Definition of Laplace Transform

Laplace Transform Pair

$$\begin{cases} F(s) = L\left[f(t)\right] = \int_{-\infty}^{\infty} f(t) e^{-st} dt \\ f(t) = L^{-1}\left[F(s)\right] = \dfrac{1}{2\pi j}\int_{\sigma - j\infty}^{\sigma + j\infty} F(s) e^{st} ds \end{cases}$$

bilateral Laplace transform

Inverse transform

$$f(t) \xleftrightarrow{LT} F(s)$$

$f(t)$--the original function; $F(s)$--the image function.

Considering that actual signals are all causal signals:

So $$F(s) = \int_{0_-}^{\infty} f(t) e^{-st} dt$$

unilateral Laplace transform

Because practical systems are **causal**, signals always start acting on the system at a certain moment, and we can regard this moment as t=0； the lower limit of the integral in the above definition is taken as 0_-， and this definition is also valid for impulse signals.

In this chapter and subsequent chapters, unless otherwise specified, it refers to the **unilateral Laplace transform**.

4.2 Definition of Laplace Transform

Fourier Transform vs. Laplace Transform

Fourier Transform	Laplace Transform
The independent variable ω is a **real number** with a clear physical meaning – **frequency**.	The independent variable $s=\sigma+j\omega$ is a **complex number**, its physical meaning is unclear and it is usually called **complex frequency**.
The Fourier transform of a signal reflects the amplitude and initial phase of different frequency components, i.e., the **spectrum** of the signal.	The Laplace transform of a signal has **no clear physical meaning**.
The Fourier transform of the system's unit impulse response is called the system's **frequency response**, which represents the characteristic that the amplitude and phase of the system output change with the input frequency when cosine signals of different frequencies act on the system.	The Laplace transform of the system's unit impulse response is called the system's **system function**. Although it is abstract, it plays an important role in characterizing system properties and system analysis.
Mainly applied to **frequency analysis** of signals and systems, such as spectrum analysis of modulation, filtering, sampling, etc.	Mainly applied to **solving differential equations**, **analysis of system functions and their zero-pole plots**, etc.

Laplace Transforms of Common Signals

Time-domain signal (t>0)	S-domain signal
$u(t)$	$\frac{1}{s}$
$e^{-\alpha t}$	$\frac{1}{s+\alpha}$
$\delta(t)$	1
$\delta(t-t_0)$	e^{-st_0}
t^n	$\frac{n!}{s^{n+1}}$

Summary

Property	Time-domain	Complex Frequency Domain
Linearity	$\sum_{i=1}^{N} a_i f_i(t)$	$\sum_{i=1}^{N} a_i F_i(s)$
Time-Delay	$f(t-t_0)u(t-t_0)$	$e^{-st_0}F(s)$
Frequency-Shifting	$f(t)e^{-at}$	$F(s+a)$
Time-Scaling	$f(at) \quad (a>0)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time-Domain Convolution Theorem	$f_1(t)*f_2(t)$	$F_1(s)\cdot F_2(s)$
Frequency-Domain Convolution Theorem	$f_1(t)\cdot f_2(t)$	$F_1(s)*F_2(s)/(2\pi j)$

Note: All the properties apply to **causal signals** only, i.e., the **signals $f(t)$ that satisfy $f(t)=f(t)u(t)$.**

Summary

Property	Time-domain	Complex Frequency Domain
Time-domain Differentiation	$\frac{df(t)}{dt}$	$SF(s)-f(0^-)$
Time-domain Integration	$\int_{-\infty}^{t} f(\tau)d\tau$	$\frac{F(s)}{s}+\frac{f^{-1}(0^-)}{s}$
s-domain Differentiation	$-tf(t)$	$\frac{dF(s)}{ds}$
s-domain Integration	$f(t)/t$	$\int_{s}^{\infty} F(\lambda)d\lambda$
Initial Value Theorem	$\lim_{t\to 0^+} f(t)=f(0^+)=\lim_{s\to\infty} sF(s)$ If **$f(t)$** contains an impulse function **$k\delta(t)$**, $\lim_{t\to 0_+} f(t)=\lim_{s\to\infty} s[F(s)-k]$	
Final Value Theorem	$\lim_{t\to\infty} f(t)=f(\infty)=\lim_{s\to 0} sF(s)$	

Find the Laplace transform of $e^{-\alpha t}\sin(\omega_0 t)u(t)$.

A $\dfrac{\omega_0+\alpha}{s^2+\omega_0^2}$

B $\dfrac{s+\alpha}{s^2+\omega_0^2}$

C $\dfrac{\omega_0}{(s+\alpha)^2+\omega_0^2}$

D $\dfrac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$

提交

Find the Laplace transform of $e^{-\alpha t}\sin(\omega_0 t)u(t)$.

Solution: $\sin(\omega_0 t)u(t)=\dfrac{1}{2j}\left(e^{j\omega_0 t}-e^{-j\omega_0 t}\right)u(t)$

$$e^{j\omega_0 t}u(t)\xleftrightarrow{LT}\frac{1}{s-j\omega_0} \qquad e^{-j\omega_0 t}u(t)\xleftrightarrow{LT}\frac{1}{s+j\omega_0}$$

$$\frac{1}{2j}\left(\frac{1}{s-j\omega_0}-\frac{1}{s+j\omega_0}\right)=\frac{\omega_0}{s^2+\omega_0^2}$$

$$\therefore \quad \sin(\omega_0 t)u(t)\xleftrightarrow{LT}\frac{\omega_0}{s^2+\omega_0^2}$$

Using the frequency shift property,

$$e^{-\alpha t}\sin(\omega_0 t)u(t)\xleftrightarrow{LT}\frac{\omega_0}{(s+\alpha)^2+\omega_0^2}$$

Contents of This Lecture

Learning Objectives of This Lecture

1. Proficiency in finding the **inverse Laplace transform** using the **partial fraction method**;
2. Proficiency in analyzing **circuits** and other **systems** using the **inverse Laplace transform**.

4.5.1 Zeros and Poles of Laplace Transform

$$f(t)=\mathcal{L}^{-1}\{F(s)\}=\frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty}F(s)e^{st}ds$$

According to the formula of the inverse Laplace transform, contour integration can be applied to solve it. However, in practice, most Laplace transforms are in the form of **rational fractions**:

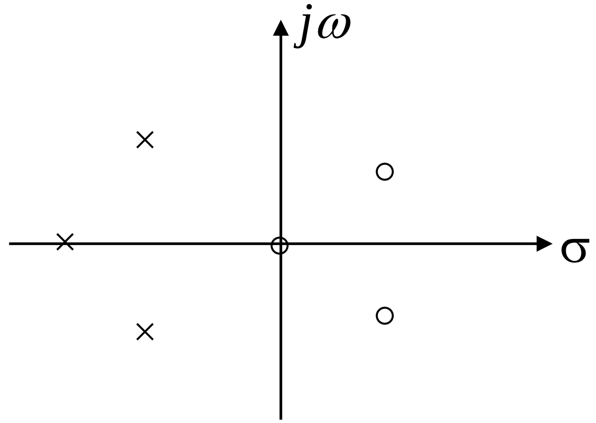
$$F(s)=\frac{N(s)}{D(s)}=\frac{\sum_{i=0}^{m}b_i s^i}{\sum_{i=0}^{n}a_i s^i}$$

Both the numerator and the denominator are **rational polynomials** in ***s*****.**

The roots of the numerator polynomial $N(s)=0$ above are called the **zeros** of ***F(s)***; the roots of the denominator polynomial $D(s)=0$ are called the **poles** of ***F(s)***. The above formula can be expressed as:

$$F(s)=\frac{N(s)}{D(s)}=\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}$$

In the formula, $\mathbf{z}_i$ and $\boldsymbol{p}_i$ are the zeros and poles of ***F(s)***, respectively. The zeros and poles can be marked on the s-plane as shown in the figure.



4.5.2 Finding the Inverse Laplace Transform

When analyzing systems using the Laplace transform method, the final step is to perform the **inverse Laplace transform** on the image function.

1. Partial Fraction Method (Heaviside Method)

(1) The Case of Simple Real Poles? $(p_1 \cdots p_n)$

When $m < n$, $F(s) \overset{Decomposition}{==} \dfrac{k_1}{s-p_1} + \cdots + \dfrac{k_n}{s-p_n}$ **(**Proper Fraction**)**

where $k_i = (s-p_i)F(s)\big|_{s=p_i}$ (Residue) (See the textbook for the proof)

It is known from the properties of Laplace transform：

$$Ae^{-\alpha t}u(t) \overset{LT}{\longleftrightarrow} \frac{A}{s+\alpha}$$

$$\boxed{F(s) \xrightarrow{L^{-1}} f(t) = \left(k_1 e^{p_1 t} + \cdots + k_n e^{p_n t}\right)u(t)}$$

Given $F(s) = \frac{s+3}{s^2+3s+2}$, then the Inverse Laplace Transform of $f(t)$ is ( ).

A $f(t) = (2e^{-t} + e^{-2t})u(t)$

B $f(t) = (2e^{t} - e^{2t})u(t)$

C $f(t) = (2e^{-t} - e^{-2t})u(t)$

D $f(t) = (e^{-t} + 2e^{-2t})u(t)$

提交

Example 4-15: $F(s) = \dfrac{s+3}{s^2+3s+2}$ ， find its Inverse Laplace Transform ***f***(***t***).

Solution: $F(s) = \dfrac{s+3}{(s+1)(s+2)} = \dfrac{k_1}{(s+1)} + \dfrac{k_2}{(s+2)}$

$$k_1 = (s+1)F(s)\,|_{s=-1} = \left.\frac{s+3}{s+2}\right|_{s=-1} = 2$$

$$k_2 = (s+2)F(s)\,|_{s=-2} = \left.\frac{s+3}{s+1}\right|_{s=-2} = -1$$

So: $F(s) = \dfrac{2}{(s+1)} - \dfrac{1}{(s+2)}$

Its Inverse Laplace Transform： $f(t) = (2e^{-t} - e^{-2t})u(t)$

Even if ***F*(*s*)** is not a proper fraction, i.e. ***n*<*m*,** it can also be expressed as the sum of a polynomial in ***s*** and a **proper fraction**:

Example:

$$F(s)=\frac{s^3+4s^2+6s+5}{s^2+3s+2}$$

Perform polynomial long division to extract the high-degree term:

$$\begin{array}{r|l} & \quad\quad\quad s+1 \\ \hline s^2+3s+2 & s^3+4s^2+6s+5 \\ & \underline{s^3+3s^2+2s} \\ & \quad\quad s^2+4s+5 \\ & \quad\quad \underline{s^2+3s+2} \\ & \quad\quad\quad\quad s+3 \end{array}$$

$$F(s)=\frac{s^3+4s^2+6s+5}{s^2+3s+2}=s+1+\frac{s+3}{s^2+3s+2}$$

The inverse Laplace transform of ***s*+1** in the above equation is:

$$\delta'(t)+\delta(t)$$

For the inverse transform of a **rational proper fraction** in Laplace transform, solve it using the **partial fraction method**.

(2) The case where poles include conjugate complex roots $(p_{1,2}=-\alpha \pm j\beta)$

$$F(s)=\frac{A(s)}{D(s)\left[(s+\alpha)^2+\beta^2\right]}$$

$$=\frac{A(s)}{D(s)(s+\alpha-j\beta)(s+\alpha+j\beta)}$$

where $D(s)$ is the remaining part of the denominator after removing the conjugate complex roots.

Set $F_1(s)=\dfrac{A(s)}{D(s)}$

then $F(s)=\dfrac{F_1(s)}{(s+\alpha-j\beta)(s+\alpha+j\beta)}$

$$\overset{Decomposition}{=\!=}\ \frac{K_1}{s+\alpha-j\beta}+\frac{K_2}{s+\alpha+j\beta}+\ldots$$

where $K_{1,2} = \dfrac{F_1(-\alpha \pm j\beta)}{\pm 2j\beta}$ (residue)

K_1、K_2 have a conjugate relationship: $K_{1,2} = A \pm jB$

Thus, the inverse transform of the part related to the conjugate complex roots is $\xrightarrow{\text{LT}^{-1}}$

$$f_c(t) = e^{-\alpha t}(K_1 e^{j\beta t} + K_1^* e^{-j\beta t})u(t)$$

$$= e^{-\alpha t}\{(A + jB)[\cos(\beta t) + j\sin(\beta t)] + (A - jB)[\cos(\beta t) - j\sin(\beta t)]\}u(t)$$

$$= 2e^{-\alpha t}\left[A\cos(\beta t) - B\sin(\beta t)\right]u(t)$$

α--attenuation factor; β--oscillation frequency

Example 4-16: $F(s)=\dfrac{s^2+3}{(s+2)(s^2+2s+5)}$, find its Inverse Laplace Transform ***f*(*t*)**.

Solution:

Let $F(s)=\dfrac{K_1}{s+2}+\dfrac{K_2}{s+1-j2}+\dfrac{K_2^*}{s+1+j2}$

$\alpha=1,$

$\beta=2,$ let $\beta>0$

$$K_1=(s+2)F(s)|_{s=-2}=\left.\frac{s^2+3}{s^2+2s+5}\right|_{s=-2}=\frac{7}{5}$$

$$K_2=(s+1-j2)F(s)\Big|_{s=-1+j2}=\left.\frac{s^2+3}{(s+2)(s+1+j2)}\right|_{s=-1+j2}=-\frac{1}{5}+j\frac{2}{5}$$

$A=-\dfrac{1}{5},B=\dfrac{2}{5}$

The time-domain signal corresponding to the conjugate complex roots is:

$$f_c(t)=2e^{-\alpha t}\left[A\cos(\beta t)-B\sin(\beta t)\right]u(t)$$

$$f(t)=\left[\frac{7}{5}e^{-2t}-\frac{2}{5}e^{-t}(\cos 2t+2\sin 2t)\right]u(t)$$

Example 4-17: $F(s)=\dfrac{s+\gamma}{(s+\alpha)^2+\beta^2}$, find its Inverse Laplace Transform ***f*(*t*).**

Solution: This function **only has conjugate complex roots**, so there is **no need to use the partial fraction method**.

known $\mathrm{LT}[\cos(\beta t)u(t)]=\dfrac{s}{s^2+\beta^2}$, $\mathrm{LT}[\sin(\beta t)u(t)]=\dfrac{\beta}{s^2+\beta^2}$

$$\mathrm{LT}[e^{-\alpha t}\cos(\beta t)u(t)]=\frac{s+\alpha}{(s+\alpha)^2+\beta^2},\quad \mathrm{LT}[e^{-\alpha t}\sin(\beta t)u(t)]=\frac{\beta}{(s+\alpha)^2+\beta^2}$$

$$F(s)=\frac{s+\alpha+\gamma-\alpha}{(s+\alpha)^2+\beta^2}=\frac{s+\alpha}{(s+\alpha)^2+\beta^2}+\frac{(\gamma-\alpha)}{\beta}\cdot\frac{\beta}{(s+\alpha)^2+\beta^2}$$

$$f(t)=e^{-\alpha t}\left[\cos(\beta t)+\frac{\gamma-\alpha}{\beta}\sin(\beta t)\right]u(t)$$

(3) The case where poles include multiple roots(a k-fold p_1, simple roots $p_2 \dots p_n$)

$$F(s) = \frac{A(s)}{D(s)} = \frac{A(s)}{(s-p_1)^k D_1(s)}$$

$$= \frac{K_{11}}{(s-p_1)^k} + \frac{K_{12}}{(s-p_1)^{k-1}} + \cdots + \frac{K_{1k}}{(s-p_1)} + \sum_{i=2}^{n} \frac{K_i}{s-p_i}$$

$$= \sum_{i=1}^{k} \frac{K_{1i}}{(s-p_1)^{k-i+1}} + \sum_{i=2}^{n} \frac{K_i}{s-p_i}$$

To find K_i and K_{11}, the method is the same as the first case:

$$K_{11} = (s-p_1)^k F(s)\Big|_{s=p_1}$$

To find other coefficients,

Let $F_1(s) = (s-p_1)^k \cdot F(s)$

$$\text{when } i=2, \quad K_{12} = \frac{\mathrm{d}}{\mathrm{d}s} F_1(s)\Big|_{s=p_1}$$

$$\text{when } i=3, \quad K_{13} = \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}s^2} F_1(s)\Big|_{s=p_1}$$

$$K_{1i} = \frac{1}{(i-1)!}\frac{\mathrm{d}^{i-1}}{\mathrm{d}\,s^{i-1}}F_1(s)\bigg|_{s=p_1} \qquad (i=1,2,\cdots k)$$

$$F_1(s) = (s-p_1)^k \cdot F(s)$$

The inverse transform of the multiple root part $\xrightarrow{\mathrm{LT}^{-1}}$:

$$f_c(t) = e^{p_1 t}\left[\frac{K_{11}}{(k-1)!}t^{k-1} + \frac{K_{12}}{(k-2)!}t^{k-2}\cdots + \frac{K_{1i}}{(k-i)!}t^{k-i}\cdots + K_{1k}\right]u(t)$$

4.5 Inverse Laplace Transform

Example 4-18: $F(s)=\dfrac{s-2}{s(s+1)^3}$, find its Inverse Laplace Transform ***f*(*t*)**.

Solution: Let

$$F(s)=\frac{K_{11}}{(s+1)^3}+\frac{K_{12}}{(s+1)^2}+\frac{K_{13}}{(s+1)}+\frac{K_2}{s}$$

First, find the coefficient K_2 corresponding to the simple root 0:

$$K_2=sF(s)\big|_{s=0}=\left.\frac{s-2}{(s+1)^3}\right|_{s=0}=-2$$

Let $F_1(s)=(s+1)^3F(s)=\dfrac{s-2}{s}$

$$K_{11} = F_1(s)\big|_{s=-1} = \left.\frac{s-2}{s}\right|_{s=-1} = 3$$

$$K_{12} = \frac{1}{(3-2)!}\left.\frac{dF_1(s)}{ds}\right|_{s=-1} = \left.\frac{d}{ds}\left[\frac{s-2}{s}\right]\right|_{s=-1} = \left.\frac{2}{s^2}\right|_{s=-1} = 2$$

$$K_{13} = \frac{1}{(3-1)!}\left.\frac{d^2F_1(s)}{ds^2}\right|_{s=-1} = \left.\frac{1}{2}\cdot\frac{-4}{s^3}\right|_{s=-1} = 2$$

$$f(t) = \left[e^{-t}\left(\frac{K_{11}}{2!}t^2 + K_{12}t + K_{13}\right) + K_2\right]u(t)$$

$$= \left[e^{-t}\left(\frac{3}{2}t^2 + 2t + 2\right) - 2\right]u(t)$$

Special Case: Irrational Expression Containing $e^{-\alpha s}$

The term $e^{-\alpha s}$ does not participate in the partial fraction operation, and the time – shifting property is used when solving.

Example 4-19: Find the Inverse Laplace Transform of $\dfrac{\mathrm{e}^{-2s}}{s^2+3s+2}$.

Solution: $$\frac{\mathrm{e}^{-2s}}{s^2+3s+2}=F_1(s)\mathrm{e}^{-2s}$$

$$F_1(s)=\frac{1}{s+1}+\frac{-1}{s+2}$$

So $f_1(t)=L^{-1}\left[F_1(s)\right]=\left(\mathrm{e}^{-t}-\mathrm{e}^{-2t}\right)u(t)$

So $f(t)=f_1(t-2)=\left[\mathrm{e}^{-(t-2)}-\mathrm{e}^{-2(t-2)}\right]u(t-2)$

4.6.1 Solving Differential Equations Using Laplace Transform

Steps for Solving Differential Equations Using Laplace Transform:

(1) Take the Laplace transform of the differential equation, paying attention to applying the differentiation property;

(2) Solve the algebraic equation in the s-domain to find the Laplace transform of the response;

(3) Find the inverse Laplace transform, i.e., find the time-domain expression of the response.

Example 4-20: Given

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+4x(t) \qquad y'(0^-)=y(0^-)=1 \qquad x(t)=e^{-3t}u(t)$$

Use the s-domain analysis method to find the zero-state response $y_{zs}(t)$.

Solution:

Take the Laplace transform of both sides of the equation, without initial conditions.

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{dx(t)}{dt}+4x(t)$$

$$s^2Y_{zs}(s)+3sY_{zs}(s)+2Y_{zs}(s)=sX(s)+4X(s)$$

Rearrange the above equation to find the Laplace transform of the zero-state response.

$$(s^2+3s+2)Y_{zs}(s)=(s+4)X(s)$$

So, $Y_{zs}(s)=\dfrac{(s+4)}{(s^2+3s+2)}X(s)=H(s)X(s)$

Since $$X(s)=\frac{1}{s+3}$$

Obtain the Laplace transform of the zero-state response:

$$Y_{zs}(s)=\frac{(s+4)}{(s^2+3s+2)(s+3)}=\frac{(s+4)}{(s+1)(s+2)(s+3)}$$

$$Y_{zs}(s)=\frac{\frac{3}{2}}{s+1}-\frac{2}{s+2}+\frac{\frac{1}{2}}{s+3}$$

Find the inverse Laplace transform to obtain the **zero-state response** of the system:

$$y_{zs}(t)=(\frac{3}{2}e^{-t}-2e^{-2t}+\frac{1}{2}e^{-3t})u(t)$$

Try to find the zero-state response using the time-domain analysis method in Chapter 2 and compare the results.

4.6.2 Solving Circuits Using Laplace Transform

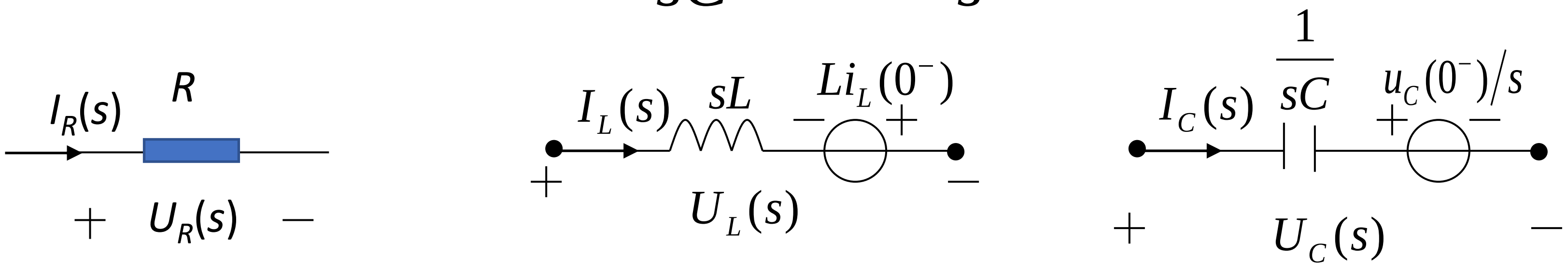
$s-$domain Component Models:

1) Loop Analysis (KVL)

$$U_R(s)=RI_R(s)$$

$$U_L(s)=sLI_L(s)-Li_L(0^-)$$

$$U_C(s)=\frac{1}{sC}I_C(s)+\frac{1}{s}u_c(0^-)$$

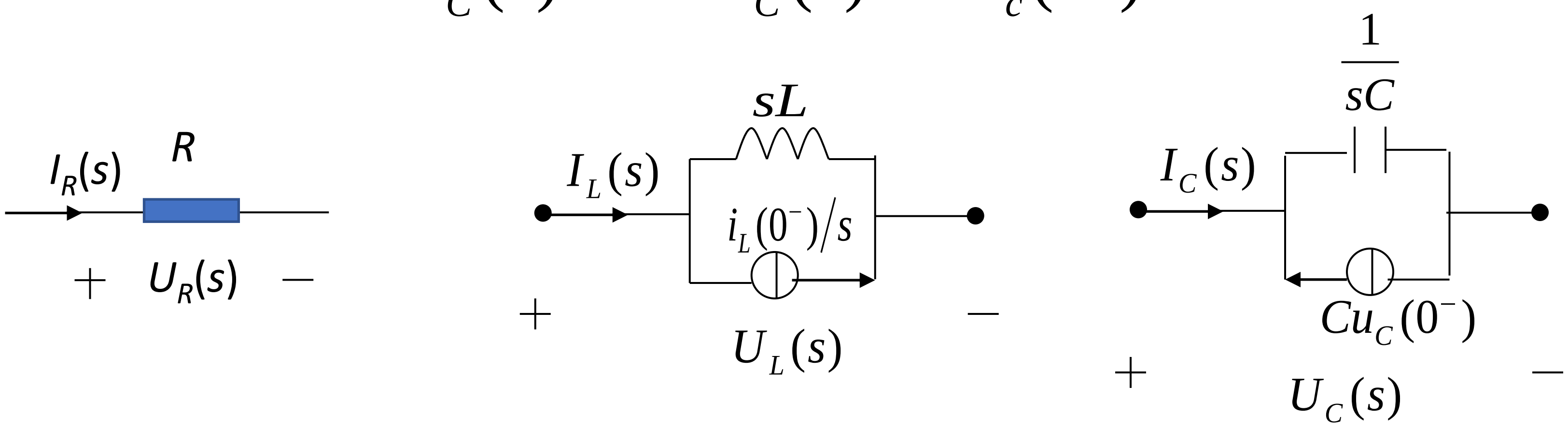


2) Node Analysis (KCL)

$$I_R = \frac{1}{R} U_R(s)$$

$$I_L(s) = \frac{1}{sL} U_L(s) + \frac{1}{s} i_L(0^-)$$

$$I_C(s) = sCU_C(s) - Cu_c(0^-)$$



s-domain Circuit Analysis Method

1. Replace each component in the network with its **s-domain model**;

Write the signal source as a transformed expression ($\frac{E}{s}$ or $\frac{E}{sR}$, $U_s(s)$ or $I_s(s)$)

2. Analyze the constructed s-domain model diagram using KVL and KCL to obtain the required transformed system equations.

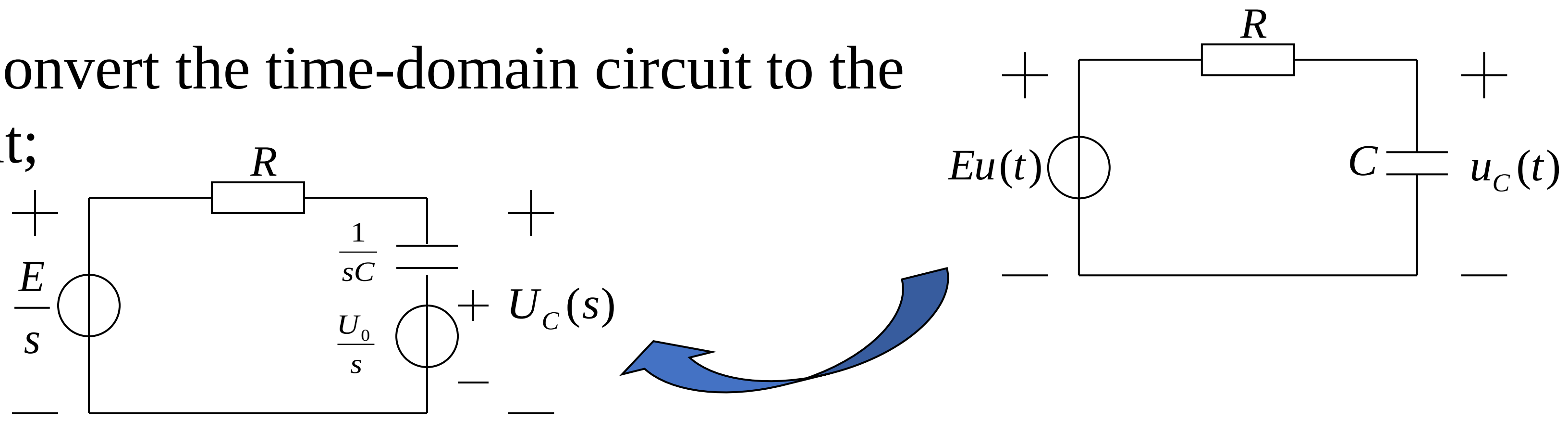
(The numerical operations performed are algebraic relationships, similar to resistive networks; Thevenin's theorem and Norton's theorem are both applicable; It is suitable for network analysis with many nodes or loops.)

3. Find the **Laplace transform** of the response.

4. Find the **inverse Laplace transform**, i.e., find the time-domain expression of the response.

Example 4-21: An RC circuit is shown in the figure. Given the initial voltage on the capacitor $\boldsymbol{u_C(0^-)=U_0}$, try to find the capacitor voltage $\boldsymbol{u_C(t)}$ for $\boldsymbol{t>0}$.

Solution: (1) Convert the time-domain circuit to the s-domain circuit;



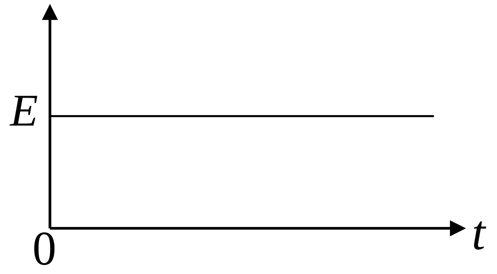
(2)List the circuit equation in the s-domain and find the Laplace transform of the output;

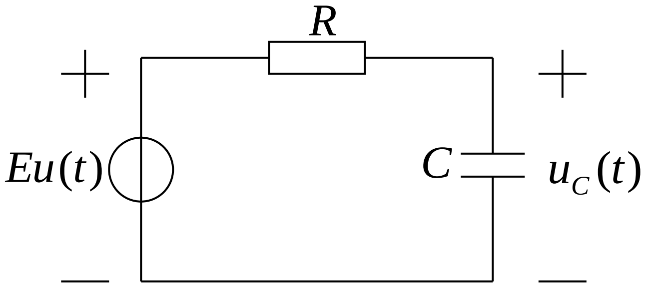
The Laplace transform of the capacitor voltage can be directly written as:

$$U_C(s)=\frac{\frac{1}{sC}}{R+\frac{1}{sC}}\frac{E-U_0}{s}+\frac{U_0}{s}=\frac{1}{sRC+1}\frac{E-U_0}{s}+\frac{U_0}{s}$$

(3)Find the time-domain expression of the response by inverse Laplace transform.

$$U_C(s)=\frac{\frac{1}{RC}(E-U_0)}{s(s+\frac{1}{RC})}+\frac{U_0}{s}=\frac{E-U_0}{s}-\frac{(E-U_0)}{s+\frac{1}{RC}}+\frac{U_0}{s}=\frac{E}{s}-\frac{(E-U_0)}{s+\frac{1}{RC}}$$



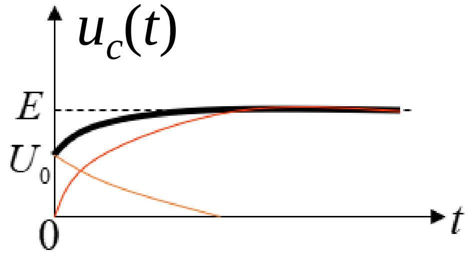


The voltage across the capacitor is obtained.

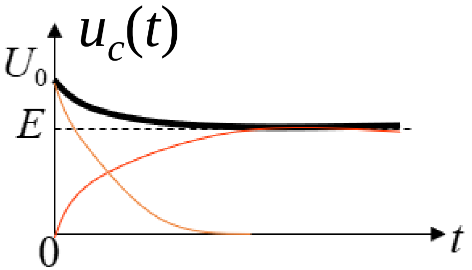
$$u_C(t)=[E-(E-U_0)e^{-\frac{t}{RC}}]u(t)$$ (black wire)

$$u_{Czs}(t)=E(1-e^{-\frac{t}{RC}})u(t)$$ (red wire)

$$u_{Czi}(t)=U_0e^{-\frac{t}{RC}}u(t)$$ (orange wire)

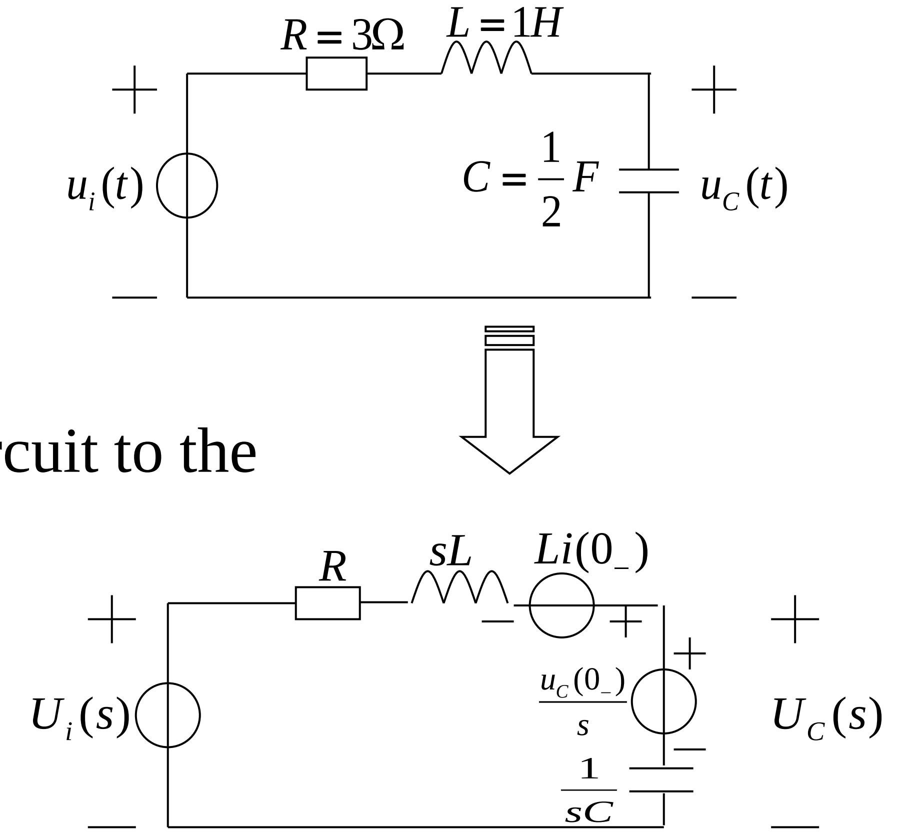


$E>U_0$



$E<U_0$

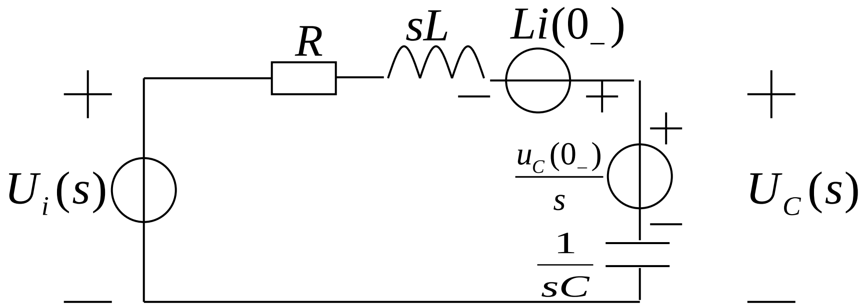
Example 4-22： An RLC series circuit is shown in the figure. Given the initial voltage on the capacitor $\boldsymbol{u_C(0_-)=1}$ **V**, the initial current in the loop $\boldsymbol{i(0_-)=1}$ **A**, and the input voltage $\boldsymbol{u_i(t)=e^{-3t}u(t)}$. Try to find the capacitor voltage $\boldsymbol{u_C(t)}$ for $\boldsymbol{t>0}$.



Solution：

(1) Convert the time-domain circuit to the s-domain circuit.

(2) List the circuit equation in the s-domain and find the Laplace transform of the response.



$$U_C(s)=\frac{\frac{1}{sC}}{R+sL+\frac{1}{sC}}[U_i(s)-\frac{u_C(0\_)}{s}+Li(0\_)]+\frac{u_C(0\_)}{s}$$

$$=\frac{\frac{2}{s}}{3+s+\frac{2}{s}}[U_i(s)-\frac{1}{s}+1]+\frac{1}{s}$$

$$=\frac{2U_i(s)}{s^2+3s+2}-\frac{2}{s^2+3s+2}(\frac{1-s}{s})+\frac{1}{s}$$

$$U_{Czs}(s) = \frac{2}{s^2+3s+2}U_i(s) = \frac{2}{s^2+3s+2}\cdot\frac{1}{s+3}$$

$$= \frac{2}{(s+1)(s+2)(s+3)} = \frac{1}{s+1} - \frac{2}{s+2} + \frac{1}{s+3}$$

$$U_{Czi}(s) = \frac{2}{s^2+3s+2}\left(\frac{s-1}{s}\right) + \frac{1}{s} = \frac{2(s-1)}{s(s+1)(s+2)} + \frac{1}{s}$$

$$= \frac{-1}{s} + \frac{4}{s+1} - \frac{3}{s+2} + \frac{1}{s} = \frac{4}{s+1} - \frac{3}{s+2}$$

(3) Find the time-domain expression of the response by inverse Laplace transform.

$$u_{Czs}(t) = (e^{-t} - 2e^{-2t} + e^{-3t})u(t) \qquad u_{Czi}(t) = (4e^{-t} - 3e^{-2t})u(t)$$

$$u_C(t) = u_{C_{zs}}(t) + u_{C_{zi}}(t) = (5e^{-t} - 5e^{-2t} + e^{-3t})u(t)$$

Homework

- Repeat the examples in the lecture notes.
- 6.37(a)(c), and 6.39(a) in the textbook.